

Haemolytic anaemia

Aplastic anaemia

Inherited causes of anaemia

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Haemolytic anaemia

- Defined as those anaemias which result from an increase in the rate of destruction of circulating red cells.
- Compensatory increase in production from bone marrow.

Classification:

- Hereditary vs acquired
- Intrinsic vs extrinsic

Haemolytic anaemia

Inherited

- **Membrane**

Hereditary spherocytosis

- **Metabolism**

G6PD deficiency

pyruvate kinase deficiency

- **Haemoglobin**

HbS, HbH, unstable

Acquired

- **Intrinsic red cell defect**

Paroxysmal nocturnal haemoglobinuria

(*PIGA* gene mutation → deficiency of glycosphosphatidylinositol → deficiency of CD55, CD59 → complement mediated lysis)

- **Extrinsic factors causing red cell damage**

1. Immune

- i. **Autoimmune**

- ii. Alloimmune

- a. Transfusion reactions

- b. Haemolytic disease of the new born

- iii. **Drug-induced immune**

2. Heat, venoms and chemicals damaging the red cell membrane

3. Drugs and chemicals oxidizing haemoglobin and other cellular components

4. Red cell fragmentation syndromes

- i. Microangiopathic haemolytic anaemia (TTP)

- ii. Mechanical haemolytic anaemia

5. Induced by micro-organisms

- i. Caused by bacterial infection of red cells (e.g. *C. perfringens*)

- ii. Caused by parasitic infection of red cells (e.g. malaria)

Haemolytic anaemia

Clinical features

- Family history
- Ethnicity
- Infection
- Drugs
- Transfusion history
- Co-existing illnesses such as autoimmune diseases and lymphoproliferative neoplasm
- History of gall-stone
- Pallor
- Jaundice – no tea-colour urine as in obstructive jaundice
- Haemoglobinuria in intravascular haemolysis e.g. PNH
- Splenomegaly

Haemolytic anaemia

Laboratory features

- Anaemia (mildly macrocytic usually) with reticulocytosis
- Increase in unconjugated bilirubin, LDH
- Reduced serum haptoglobin
- Increased in methaemalbumin

Blood film

- Polychromasia
 - Spherocytes (hereditary spherocytosis, immune haemolytic anaemia)
 - RBC fragmentation (microangiopathic haemolysis)
 - RBC agglutination (cold agglutinin disease)
-
- Direct antiglobulin test – positive in immune haemolytic anaemia

Autoimmune haemolytic anaemia (AIHA)

- Due to autoimmune antibodies against self erythrocytes
- Warm or cold auto-antibodies depending on the temperature with which the antibody binds better with erythrocytes (37°C vs 4°C)

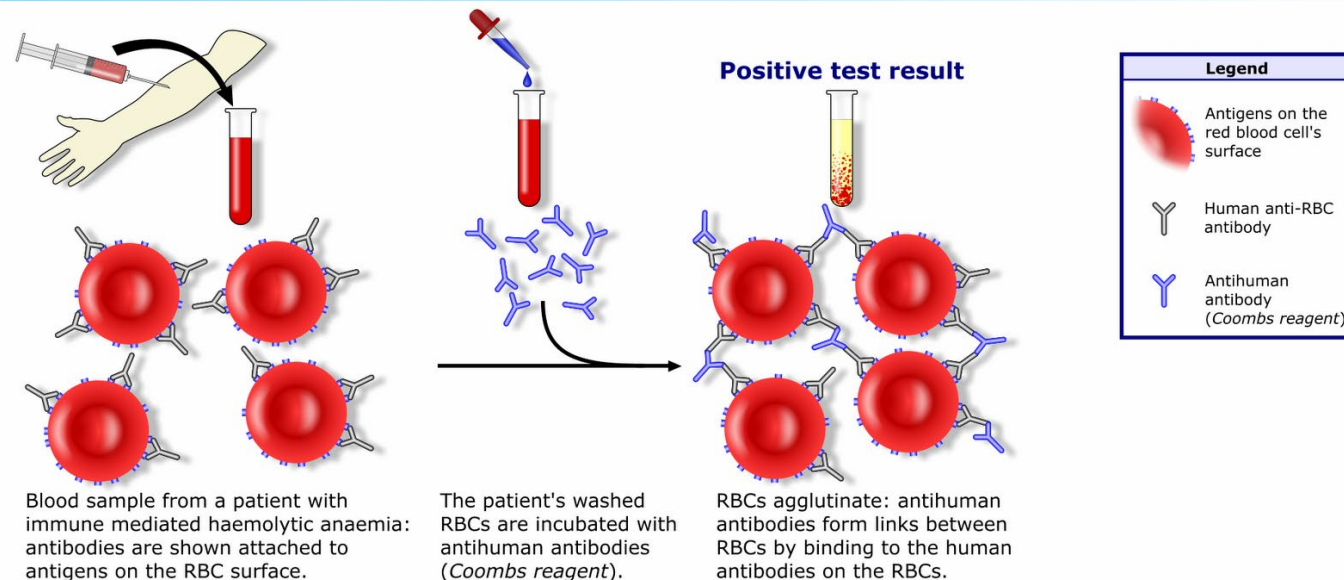
Warm type Often IgG	Cold type Often IgM
Idiopathic	Idiopathic – cold haemagglutinin disease
Secondary • Drugs e.g. methyldopa • Autoimmune diseases e.g. SLE • Lymphoproliferative disorders e.g. CLL	Secondary • Infections e.g. mycoplasma pneumonia, infectious mononucleosis • Lymphoproliferative disorders

AIHA

diagnosis

- Blood count- increase in **reticulocyte**
- **Blood film** - microspherocytes
- Increased in **bilirubin (unconjugated)**, LDH
- Decreased in hepatoglobin, increase in methaemalbumin
- Specific tests - direct antiglobulin test (direct Coombs' test)

Direct Coombs test / Direct antiglobulin test



AIHA

treatment

1. Transfusion if necessary
2. Folate supplement
3. Decrease further haemolysis, e.g.
 - Steroid or other immunosuppressants in autoimmune haemolytic anaemia
 - Avoidance of certain drugs in drug-induced haemolytic anaemia
 - Keep warm in cold haemagglutinin disease
 - Splenectomy
4. Treat the underlying cause (e.g. AIHA due to underlying lymphoproliferative disorder) if necessary

Aplastic anaemia

- Pancytopenia resulting from bone marrow hypoplasia or aplasia
- Congenital vs acquired
- Acquired: idiopathic vs secondary
- Isolated aplasia of the erythroid series (pure red cell aplasia; PRA)

Congenital: Diamond-Blackfan syndrome

Acquired: lymphoproliferative disease, thymoma, parovirus B19

Etiology

- Idiopathic – 70-80%
- Congenital
 - Fanconi anaemia, Shwachman-Diamond syndrome, Dyskeratosis congenita
- Drugs
 - Gold, Alcohol, Diclofenic acid, Indomethacin, Chloramphenicol, anti-convulsants
- Infection
 - non A,B,C hepatitis (sero-negative hepatitis)
- Environmental exposure
 - benzene, pesticide

Idiopathic aplastic anaemia

Pathogenesis

- Immune (T-cell) mediated suppression of marrow stem cells

Aplastic anaemia- clinical features

- Symptoms of pancytopenia (anaemia, neutropenia, thrombocytopenia)
- **NO lymphadenopathy**
- **NO hepatosplenomegaly**
- In young patients, watch out for the presence of congenital physical abnormalities (**inherited bone marrow failure syndromes**)

Aplastic anaemia- investigations

- Blood count: pancytopenia; macrocytic anaemia; **low reticulocyte count**
- Blood film
- Autoimmune markers
- Vit B12 and folate levels
- Bone marrow: trephine biopsy is required for the assessment of cellularity
- Specialized tests:
 - Flow cytometry for CD55 and CD59 deficient RBC (to rule out PNH)
 - Chromosome breakage with diepoxybutane (to screen for Fanconi anaemia)

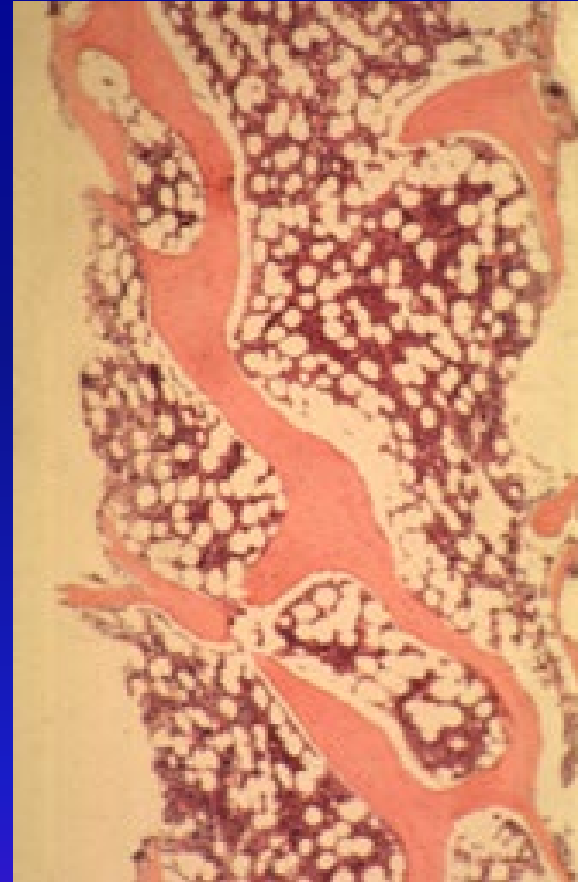
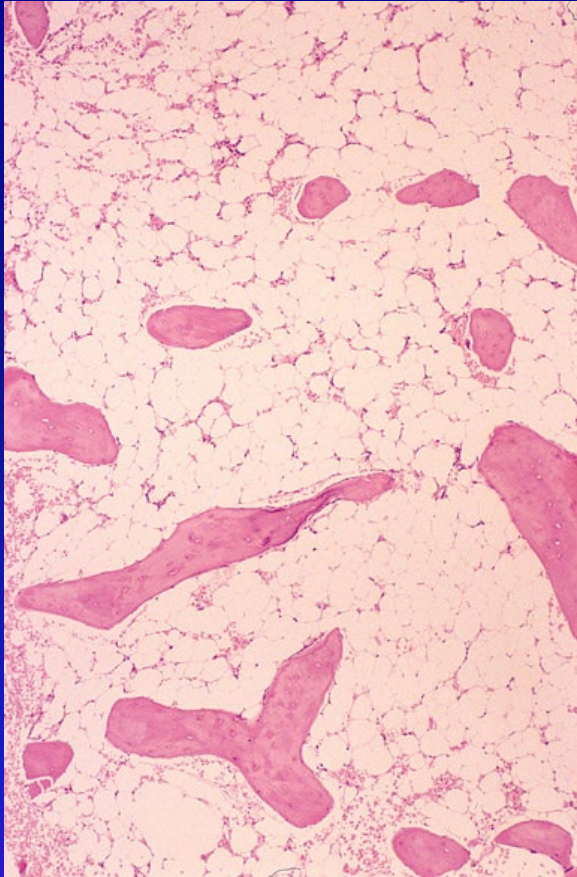
Definition of severity

1. **Severe** aplastic anaemia

- BM cellularity <25%
- Neutrophils < $0.5 \times 10^9/\text{L}$ }
- Platelets < $20 \times 10^9/\text{L}$ } 2 out of 3
- Reticulocytes < $20 \times 10^9/\text{L}$ }

2. **Very severe** aplastic anaemia

- As for severe AA but neutrophils < $0.2 \times 10^9/\text{L}$



SEVERE aplastic anaemia- treatment

- First line - allogeneic haematopoietic stem cell transplantation (for young patients with matched sibling donors)
- Anti-thymocyte globulin + cyclosporine
- Cyclosporine A
- Supportive (blood products, iron chelation)
- Eltrombopag (high dose) + ATG and cyclosporine

Inherited bone marrow failure syndromes

Pancytopenia

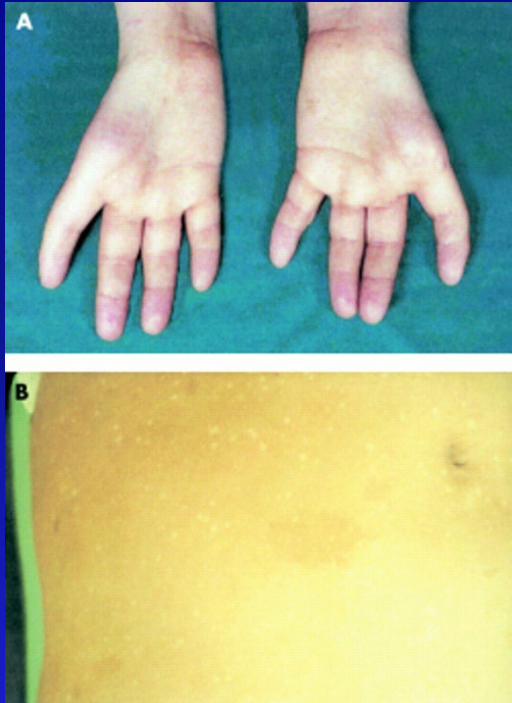
- Fanconi anaemia
- Dyskeratosis congenita
- Shwachman-Diamond syndrome

Anaemia

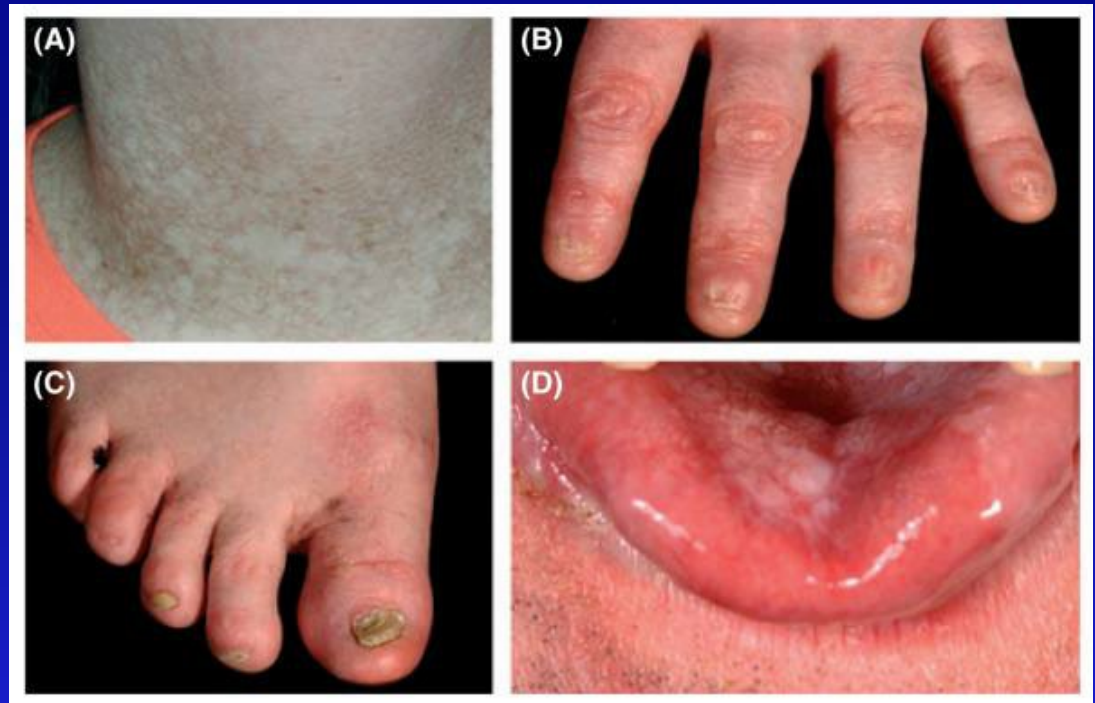
- Diamond-Blackfan anaemia
- Congenital dyserythropoietic anaemia

Inherited BM failure syndromes

	Fanconi anaemia	Dyskeratosis congenita	Shwachman-Diamond syndrome	Diamond-Blackfan syndrome	Congenital dyserythropoietic anaemia
Inheritance	AR, XLR	XLR, AR, AD	AR	AD	AR, AD
Somatic abnormalities	Yes	Yes	Yes	Yes	Rare
BM failure	Aplastic anaemia	Aplastic anaemia	Aplastic anaemia	Pure red cell aplasia	Ineffective erythropoiesis
Short telomeres	Yes	Yes	Yes	?	?
Malignancy	Yes	Yes	Yes	Yes	?No
Chromosome instability	Yes	Yes	Yes	?	?
Genes involved	13	6	1	7	1



Fanconi anaemia



Dyskeratosis congenita

Inherited causes of anaemia

Pathophysiology

- Inherited bone marrow failure syndromes - pancytopenia
- Disorders of red cell membrane
- Disorders of haemoglobin (haemoglobinopathy and thalassaemia)
- Disorders of RBC metabolism

Inherited causes of anaemia

History

- Family history – pattern of inheritance
- Age of onset
- Drug history
- Symptoms of haemolysis
- Transfusion requirement
- Complications (including those due to long-term red cell transfusion)

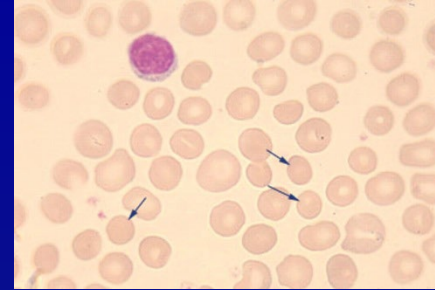
Inherited causes of anaemia

Physical examination

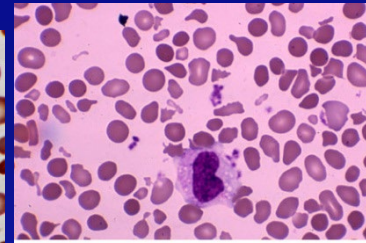
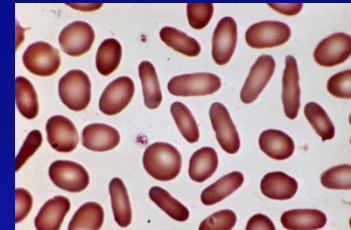
- Pallor
- Jaundice
- Splenomegaly
- Skin complexion
- Skeletal deformities

Disorders of red cell membranes

- Hereditary spherocytosis

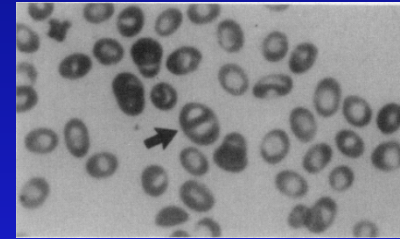


- Hereditary elliptocytosis and hereditary pyropoikilocytosis

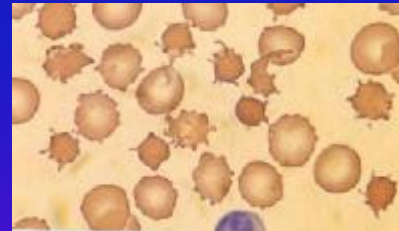


Source: Lichtman MA, Shafer MS, Felgar RE, Wang N: *Lichtman's Atlas of Hematology*: <http://www.accessmedicine.com>
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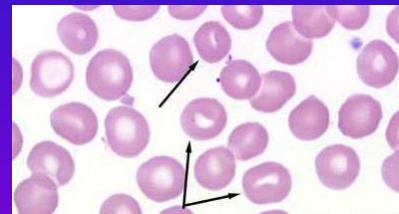
- Southeast Asian ovalocytosis



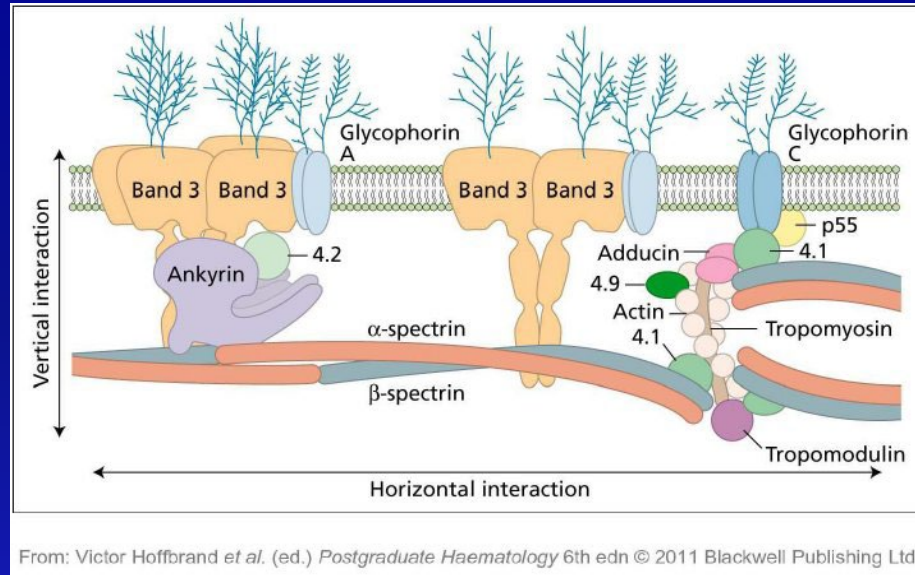
- Hereditary acanthocytosis



- Hereditary stomatocytosis



Disorders of red cell membrane



Protein

α-spectrin

β-spectrin

Ankyrin

Band 3

Protein 4.1

Protein 4.2

Function

cytoskeleton network

cytoskeleton network

Vertical contact

Anion exchange channel

Spectrin – actin contact

Spectrin – ankyrin interaction

Associated disorders

Hereditary spherocytosis, hereditary elliptocytosis

Hereditary pyropoikilocytosis, HS, HE

Hereditary spherocytosis

HS, SEA ovalocytosis, hereditary acanthocytosis

HE

HS (Japan)

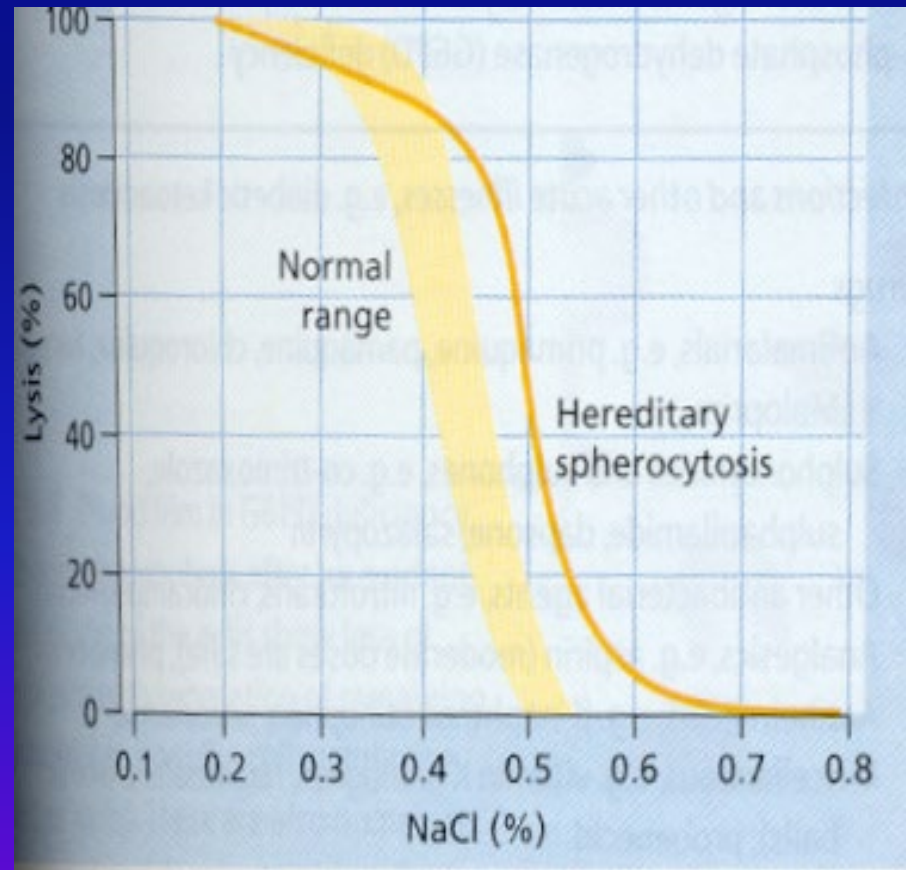
Hereditary spherocytosis

- Presence of spherocytes
- Haemolytic anaemia – mild to severe
- Autosomal dominant
- 60% result from a defect in the ankyrin – spectrin complex or vertical interaction

Hereditary spherocytosis

- Jaundice – indirect hyperbilirubinaemia
- Splenomegaly
- Gall-stone
- Aplastic crisis – primary parvovirus B19 infection
- Extravascular haemolysis
- Blood film - spherocytes
- MCHC in often raised
- Osmotic fragility test
- Eosin-5-maleimide binding (decreased binding in HS)

Hereditary spherocytosis



Hereditary spherocytosis

- Folic acid supplement
- Compensated haemolysis – no specific treatment
- Cholecystectomy if symptomatic gall-stone
- Splenectomy in severe haemolysis

Disorders of haemoglobin

Thalassaemia

- Disorders of haemoglobin with a decrease in the rate of synthesis of one or more globin chains

Haemoglobinopathies

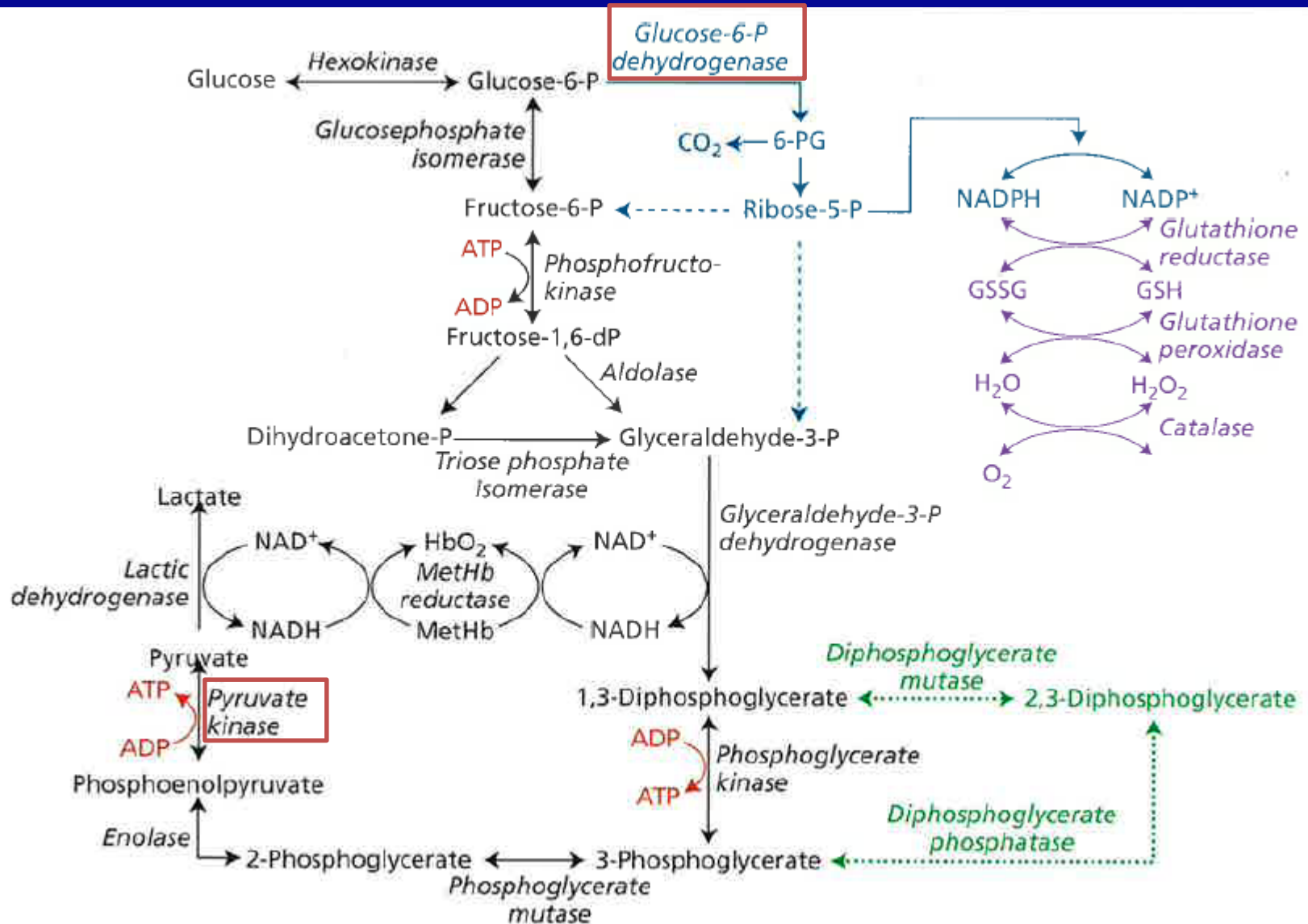
- Haemoglobin disorders with a structurally abnormal haemoglobin e.g. Hb S
- In some cases of haemoglobinopathies, there are concomitant reduced synthesis of the globin chains “thalassaemic haemoglobinopathies” e.g. Hb Constant Spring, Hb E

Please refer to your WCS on inherited disorders of haemoglobin

Disorders of red cell metabolism

- Glucose-6-phosphate dehydrogenase deficiency
- Pyruvate kinase deficiency
- Deficiency of other glycolytic pathway enzymes
- Congenital non-spherocytic haemolytic anaemia

Glycolytic pathway and pentose phosphate shunt



G6PD deficiency

- X-linked recessive
- Incidence in HK: 4.5% in males and 0.5% in females
- More than 100 variants associated with enzyme deficiency.
- Type B refers to the wild-type

G6PD deficiency

Class	Enzyme activity	Example	Clinical effects
I	Severe < 2%	Santiago de Cuba (Gly447Arg)	Haemolytic anaemia, acute exacerbation
II	<10%	- Canton (Arg45Leu) - Mediterranean (Ser188Phe)	Favism, acute intravascular haemolysis (drug induced), neonatal jaundice
III	Moderate <60%	A- (Val68Met , Asn126Asp)	None
IV	100%	- B (wild type) - A (Asn126Asp)	None

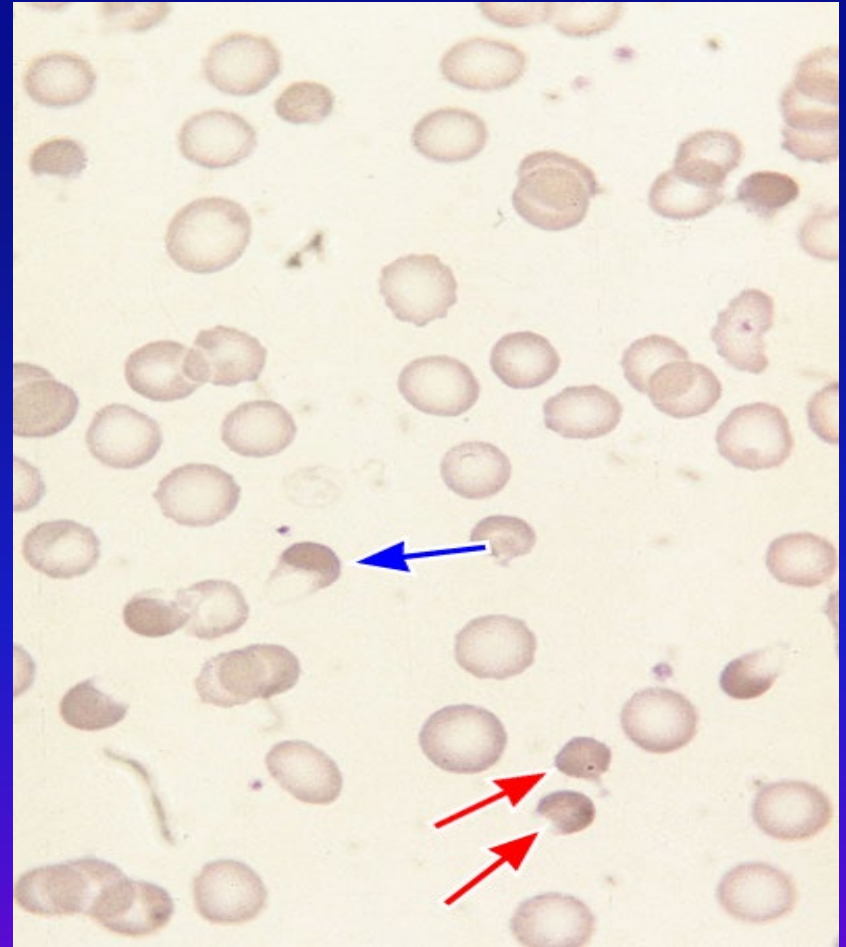
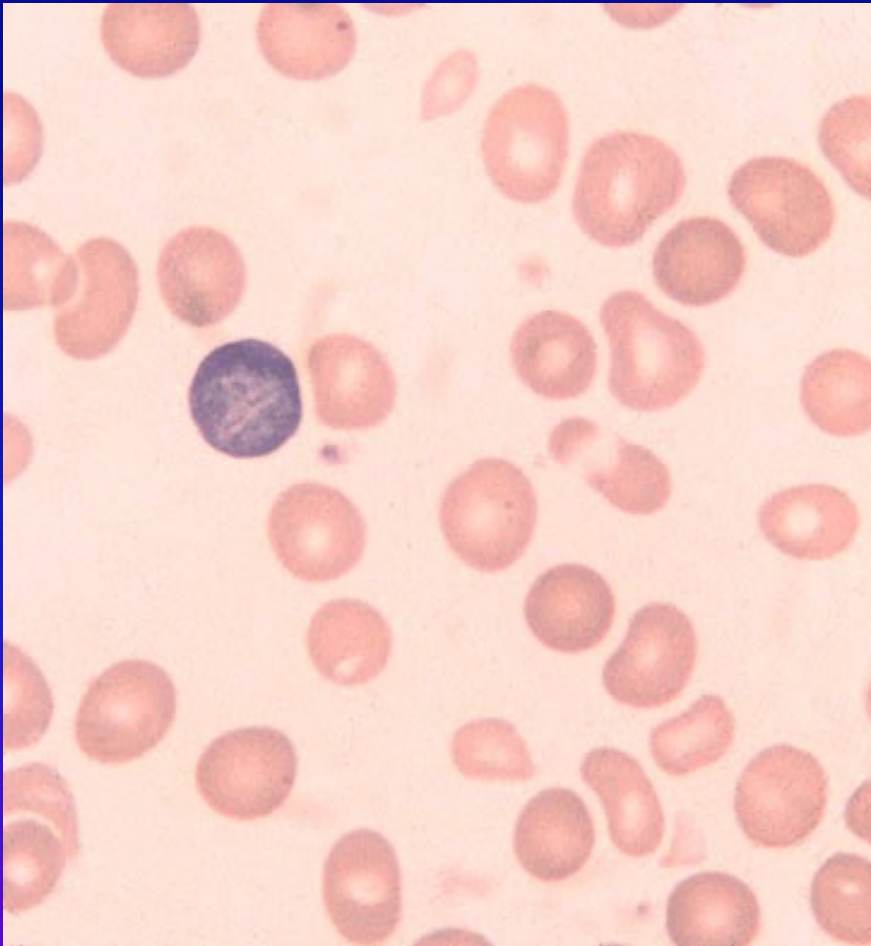
G6PD deficiency

- Intravascular haemolysis with haemoglobinuria – acute or chronic (not common)
- Precipitated by infection or acute medical illness, drugs and Fava beans
- Drugs: sulphonamides, rasburicase, primaquine

G6PD deficiency

Blood film – polychromasia, ghost or hemi-ghost cells

G6PD assay – may be normal during the acute phase as young red cells and reticulocytes may have normal G6PD levels



G6PD deficiency

- Supportive
- Blood transfusion
- Folic acid
- Stop and avoid offending agents

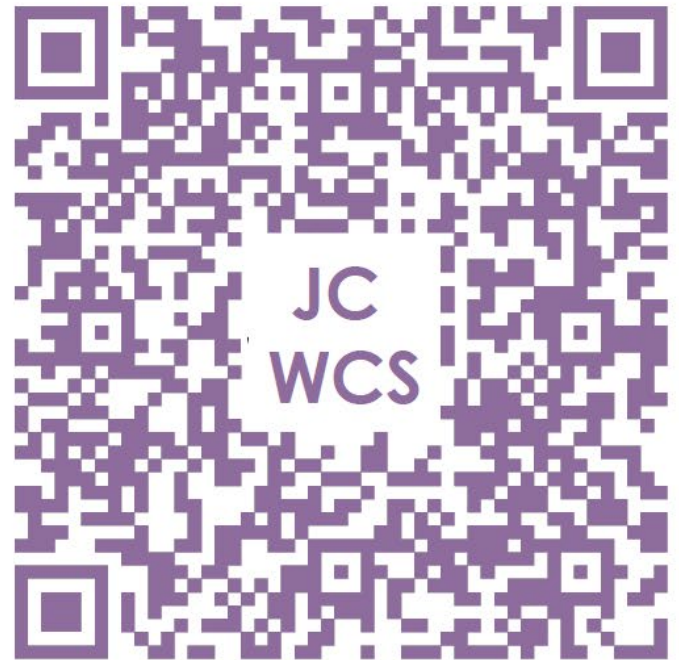
THE END

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)